Startup Dynamics: Transitioning from Nonemployer Firms to Employer Firms, Survival, and Job Creation

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Abstract

Understanding the dynamics of startup businesses' growth, exit, and survival is crucial for fostering entrepreneurship. Among the nearly 30 million registered businesses in the United States, fewer than six million have employees beyond the business owners. This research addresses the gap in understanding which companies transition to employer businesses and the mechanisms behind this process. Job creation remains a critical concern for policymakers, researchers, and advocacy groups. This study aims to illuminate the transition from non-employer businesses to employer businesses and explore job creation by new startups. Leveraging newly available microdata from the U.S. Census Bureau, we seek to gain deeper insights into firm survival, job creation by startups, and the transition from non-employer to employer status.

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Executive Summary

Understanding the dynamics of startup business survival and growth is crucial for fostering a more entrepreneurial economy. Of the nearly 30 million registered businesses, fewer than 6 million have any additional employees beyond the business owners. There has been a decline in startup rate and the share of activity accounted for by young firms, which are typically the main driver of job creation, over the last couple of decades (Haltiwanger, 2022).

Job creation remains a critical issue for policymakers, researchers, and advocacy groups. This research aims to shed light on the transition from non-employer businesses to employer businesses and job creation by new startups. Despite the significance of this transition, researchers are only beginning to have a better understanding of which non-employer companies become employer businesses and the process they undergo (Davis et al., 2007). Using the newly available Comprehensive Startup Panel (CSP) (2011-2019) and the 2012 Survey of Business Owners (SBO), we track startups that began operations in 2011 or 2012—both with and without employees—over the 2012-2019 period. This enables us to understand better the transition from non-employer to employer status, firm survival, and job creation by startups. The matched CSP-SBO sample offers a unique opportunity to examine a large set of more than 100,000 startup businesses tracked over their first seven years, providing detailed information on the firms, owners, and strategy.

We find strong evidence that the legal form is related to the likelihood of transitioning to an employer firm with corporations and partnerships more likely to transition than their sole proprietor counterparts. We also find financial capital investment is another critical factor influencing the likelihood of transition with higher levels of investment being correlated with the

likelihood to transition. Other firm characteristics that were positively associated with transitioning from nonemployer to employer were joint ownership between spouses, having a website, and having customers that were other business and/or government. Owner demographics, such as sex, race, and ethnicity, generally do not significantly influence the transition. Business owners ages 25 to 44 are more likely to transition their businesses compared with businesses with the youngest owners (under 25). Previous business experience had a positive and statistically significant effect, as did having a business that provides the primary source of income. Hours worked per week is a significant positive factor; more hours worked correlates with a higher likelihood of transitioning, emphasizing the importance of effort and dedication in moving from nonemployer to employer status.

Regarding business survival, nonemployer startups are less likely than employer startups to survive. Legal form strongly correlates with survival: corporations, partnerships, and other legal forms are more likely to survive than sole proprietorships. Higher levels of financial capital investment are associated with higher survival rates. Firm characteristics positively associated with survival include joint ownership between spouses, exporting goods or services, having e-commerce capabilities, and maintaining a website. In terms of owner characteristics, businesses owned by women were less likely to survive than businesses owned by men, controlling for other factors. Businesses owned by blacks and Native Hawaiian and Other Pacific Islanders (NHOPI) were less likely to survive than businesses owned by whites. Veteran-owned businesses were less likely to survive than businesses owned by non-veterans. Owner age, education, and hours worked were all important predictors of business survival. Businesses with owners working more hours than 20 hours a week were more likely to survive than businesses with owners

working less than 20 hours a week. Finally, businesses that were the primary source of income for the owner were more likely to survive than businesses that were not the primary source of income.

Our results also provide several key insights into factors that influence positive net employment growth thereby informing potential policy approaches around job creation. We find that access to financial capital is crucial, so programs aimed at reducing capital constraints can support job creation and business expansion. Supporting a website presence can help businesses engage with customers. Targeted support for female, Black, and veteran entrepreneurs can help overcome systemic barriers to growth. Promoting practical business experience and dedicated work effort could also support the job creation outcomes.

Our analysis of a large sample of startup businesses tracked over seven years post-startup provides key insights into factors influencing the dynamics of these businesses, including both survival and growth. As the SBA aims to be more impactful in its support of early-stage companies, we hope our findings offer some guidance on focusing efforts to be most effective.

Introduction

This research seeks to improve our understanding of the transition from non-employer businesses to employer businesses and job creation by new startups. Job creation is a critical issue for policymakers, researchers, and advocacy groups. Of the nearly 30 million registered businesses, fewer than 6 million have any additional employees beyond the business owners. We seek to provide a better understanding of which companies become employer businesses and what that process looks like.

Using confidential microdata that are available to Sworn Special Status researchers with an approved research project at the Center of Economic Studies, this research examines three different dynamics of early-stage entrepreneurship.

1) We examine nonemployer startups from 2011/2012 and track them over an eight year period to see how many of them transitioned to becoming employer firms;

2) We examine all startups (nonemployer and employer) from 2011/2012 and track them over an eight year period to see how many of them survived; and

3) We examine all startups (nonemployer and employer) from 2011/2012 and see how many of them had positive net employment growth over the eight year period. This would include nonemployer startups that transitioned to employer businesses and employer startups that added new employees (over their original base year employment) over the period of observation.

Using the Comprehensive Startup Panel, we examine the universe of 2011 and 2012 startup businesses and study the distribution of firms in sets of buckets that break out nonemployer startups and employer startups, as well as the status of these startups in terms of job creation and survival. We then examine a subsample of the data that were matched to the 2012 Survey of Business Owners and compare this sample to the full universe. Next, we delineate in a univariate analysis the distribution of these startups by firm status and by various firm and owner characteristics. Finally, we use multivariate analysis to examine the three entrepreneurial dynamics (transition, survival, and job creation) and the role that various firm and owner characteristics play in these outcomes. The final section provides a summary and conclusions.

Literature Review

Understanding the dynamics of startup businesses' growth, exit, and survival is crucial for fostering entrepreneurship (Davis et al., 2007). Several studies have explored the contribution of different types of firms to the United States economy, finding that nonemployer businesses start at rates three times that of employer firms and that high-growth firms play a key role in generating jobs (Acs & Mueller, 2008; Acs, Headd, & Agwara, 2009; Gompers et al., 2010; Fairlie & Miranda, 2017). Small businesses, including startups and nonemployers, notably contribute to job creation, especially during economic expansions, and often evolve into large employer firms (Neumark, Wall, & Zhang, 2011; Haltiwanger, Jarmin, & Miranda, 2013).

Gompers et al. (2010) and Decker et al. (2016) examine the persistence of startups in the United States. Gompers et al. (2010) find that high-performing firms are more likely to sustain growth and transition to employer status. Decker et al. (2016) note a decline in young high-growth firms over time. Similarly, Hathaway and Litan (2014) and Decker et al (2014) argue that business dynamism has been on the decline since the late 2000s, with older and larger firms faring better than younger ones, and fewer people launching new firms.

The scholarly literature on entrepreneurship and growth relationships is broad and varied. At the individual venture level, knowledge about entrepreneurial growth dynamics remains limited (Parker, 2009). Empirical studies analyzing time-series data commonly explain only a small fraction of the variation in enterprise growth rates. This limited progress may be due to the substantial heterogeneity of entrepreneurs and their ventures (Santarelli & Vivarelli, 2007) and the idiosyncratic shocks those ventures face. Another reason might be that growth itself is not very common. Research has shown there is little relationship between firm size and firm growth

conditional on firm age. Employment growth is driven by young firms, which also happen to be small (Haltiwanger et al., 2010). Pugsley and Hurst (2011) show that even among young firms and conditional on survival, growth is still rare overall. Growth is often conditional on owners' reasons for starting a business. Hurst and Pugsley (2011) report that most employer business owners create firms for nonpecuniary reasons and thus have no desire to grow, indicating pronounced implications for nonemployer businesses in terms of growth outcomes.

Focusing on the determinants of venture growth, a useful framework groups applicable determinants into three overlapping categories: founder characteristics; firm traits; and strategies employed for venture viability, growth, and success (Storey, 1994). Research has found that human capital obtained through education is one driver of entrepreneurial performance, irrespective of whether earnings, survival, employment, or sales growth are used as performance measures (Van Der Sluis et al., 2008; Unger et al., 2011). Research by Bates, Robb, and Parker for SBA's Office of Advocacy (2011) found that high-growth firms (5X revenue growth over the study period) were more likely to have owners with graduate degrees than their non-high-growth counterparts. In a multivariate setting with a competing risk model, higher education levels were shown to be related to higher growth prospects.[†] That analysis used the data from the Kauffman Firm Survey over the 2004-2008 period.

While there is consensus that a stronger founder educational background enhances firm growth (Cooper et al., 1991; Cooper et al., 1994; Reynolds & White, 1997; Almus & Nerlinger, 1999; Kangasharju & Pekkala, 2002), disagreement exists over the validity of this relationship

[†] From that Final Report in 2011: "Regardless of the growth percentile at which the threshold was set, possession of undergraduate degrees as well as graduate or professional training was found to have a significant positive impact on the likelihood that a given entrant survives and prospers after three or more years of being in business. The magnitude of this effect increases with the height of the threshold, with a fivefold increase in the impact of advanced educational credentials at 90th percentile growth performance, relative to 10th percentile growth performance."

(Haber & Reicchel, 2007; Schutjens & Wever, 2000). For more thorough reviews of the literature, please see Van Der Sluis et al. (2008) and Unger et al. (2011). Another measure of human capital, applicable work experience, is widely seen as enhancing venture growth prospects (Koeller & Lechler, 2006; Cooper et al., 1991; Cooper et al., 1994). However, other founder traits such as age, gender, growth motivation, and training show no consensus regarding their relationship to firm growth (Storey, 1994; Tamasy, 2006; Capelleras & Greene, 2008; Parker, 2009; Fairlie and Miranda, 2017; Azoulay et al., 2020).

Financial capital has also been shown to play a role in both firm creation and firm performance. Fairlie et al. (2023) suggest that adequate startup capitalization is important for predicting failure/closure due to insufficient capital and likely job creation with adequate capital. Fairlie and Robb (2008) found better capitalized businesses had higher sales, profits, and employment, and were less likely to close than businesses that received lower levels of start-up capital. Westhead and Cowling (1995) found that firms that had access to and used multiple sources of start-up finance tended to grow faster. All of these factors will be examined in our study.

The history of nonemployer firms as it relates to entrepreneurial success, job creation, and survival is often excluded from standard studies of entrepreneurship (Fairlie et al., 2023). Limited data on the universe of nonemployer startups has hindered understanding of their transition to employer status. However, Fairlie et al. (2023) investigate this transition using a Comprehensive Start-up Panel (CSP) tracking nonemployers from their initial start date and following them for seven years and find that about 20 percent of new employer firms were once nonemployers that transitioned. They also document the importance of these nonemployer-employer transitions in overall job creation: the share of total job creation by nonemployer startups

increased from six percent one year after startup to 14 percent seven years after startup[‡]. One unique study by Cockx and Desiere (2024), leveraged a unique policy in Belgium that permanently reduced the labor cost of the first employee by 13% and found that the number of new, first-time employers jumped by 31% immediately following the reform.

In summary, the literature emphasizes the critical role of high-growth firms, firm and founder characteristics, and strategic choices in entrepreneurial success. Our study aims to contribute to this body of knowledge by investigating nonemployer firms' transitions to employer status and the dynamics of firm survival and job creation leveraging novel and comprehensive data sources. Our study specifically builds on the work of Fairlie et al. (2023) by using a cohort of startup businesses that began operations in 2011 and 2012 from the Comprehensive Startup Panel and linking them to the 2012 Survey of Business Owners. These merged data allow us to investigate the role of firm characteristics, owner characteristics, and strategies on the transition of nonemployer businesses to employer businesses, business survival, and job creation through positive net employment change over the period of observation.

Data and Methodology

The data used in this research study are the Comprehensive Startup Panel (CSP) and the Survey of Business Owners (SBO), which are described in more detail below. These confidential microdata are available to Sworn Special Status researchers with an approved research

[‡] Fairlie et al (2024) found the average employer startup creates 6.5 jobs in the first year after startup, but falls to 4.6 jobs in the seventh year after startup (without conditioning on survival). The total number of jobs created by nonemployer startups steadily increases relative to the total number of jobs created by employer startups over time. They found nonemployer startups hired 181,000 employees in their first year after startup. Employer startups hired 2.9 million employees in the first follow-up year, but the differential narrowed over time. Seven years after startup, they found that nonemployer startup hiring grew to 319,000 employees and employer startup hiring dropped to 2.0 million employees.

project through the Census Bureau's Federal Statistical Research Data Center (FSRDC), which has more than 30 centers throughout the United States. Researchers interested in accessing these or other data can apply through the following link: https://www.census.gov/about/adrm/ced/apply-for-access.html.

The Comprehensive Startup Panel (CSP) provides longitudinal data on several startup cohorts (currently 1995 to 2014 startup cohorts). The panel data set follows each startup cohort for up to seven years after the startup year and is the combination of two existing Census Bureau panel data sets:

(1) the Longitudinal Business Database (LBD), which is a panel of all nonfarm employer establishments beginning in 1976, and

(2) the Integrated Longitudinal Business Database (ILBD), a longitudinal panel of nonemployer establishments, beginning in 1994 (the ILBD contains cross-sectional data and five-year longitudinal linkages from 1977 to 1992 as well).

Through combining the firm-level LBD and the ILBD, we can define all startups, nonemployer and employer, as well as track these businesses over time to examine survival and job creation. Startups are defined as the first year a firm has revenues (for nonemployers) or payroll (for employers). See Fairlie et al. (2023) for a much more detailed explanation of the dataset and how the linkages between the LBD and ILBD were made.

For this research, we make use of the 2011 and 2012 startup cohorts from the CSP and the corresponding information on startup year (2011 or 2012), a flag that is equal to one if the business is starting as a nonemployer, a flag that is equal to one if the firm transitioned from nonemployer to employer over the period, the transition year (0 indicating the firm transitioned in its

first year (2011 or 2012), 1 if it transitioned in its second year, etc.), a flag for whether or not the business survived in each year, and employment (if any) in each year[§].

The Survey of Business Owners (SBO) was a firm-level survey conducted every five years by the U.S. Census Bureau in years ending in two and seven. However, 2012 was the last year the Census conducted the survey and it has since been replaced by the Annual Business Survey (ABS). Except for businesses in a small number of specific industries, the frame for the 2012 SBO was all businesses in operation during 2012 with revenues of at least $1,000[**]. The ABS does not sample nonemployers, so the Nonemployer Statistics by Demographics series (NES-D) is now the source of information on the demographic characteristics of nonemployer businesses[††].

The SBO is one of the few sources of data that contain both firm characteristics and the characteristics of the owner(s) of the firm. This rich dataset contains detailed demographics on up to four owners of each firm, as well as other interesting and relevant owner characteristics such as education level, previous business experience, and hours worked in the business[‡‡]. Firm characteristics include legal form of organization, industry, the amount of financial capital used to start up, whether or not the firm had any intellectual property, exports, e-commerce, and/or a website. The following sections will provide more detail about the variables we use from the SBO.

[§] "Survival" in the CSP means active in a given year as observed in either the ILBD (i.e., has revenue) or LBD (i.e., has payroll).

[**] https://www.census.gov/programs-surveys/sbo/technical-documentation/methodology/2012-sbo-methodology.html

[††] See https://www.census.gov/programs-surveys/abs/data/nesd.html and https://www.census.gov/programs-surveys/abs.html for more information on the ABS and NES-D.

[‡‡] For the owner characteristics (age, education, hours worked, etc.) in the cases of multi-owner firms we use the characteristics of the primary owner designated in the SBO. We also examined using the average of these characteristics across owners and found similar results.

Initial Analysis

We first used the CSP to assign the status of each firm into a set of four buckets:

1) Those firms that started and ended as a nonemployer;

2) Those firms that started as a nonemployer but transitioned to becoming an employer over the period;

3) Those firms that started as employer firms and had no positive net job creation over the period; and

4) Those firms that started as employer firms, and had positive net job creation over the period;

Taking survival into account, firms were then assigned a status into a set of six buckets, as follows:

1) Those firms that started and ended as a nonemployer and survived over the period;

2) Those firms that started as a nonemployer but transitioned to becoming an employer over the period and survived;

3) Those firms that started as employer firms and survived but had no positive net job creation over the period;

4) Those firms that started as employer firms, survived, and had positive net job creation over the period;

5) Started as a nonemployer and did not survive; and

6) Started as an employer firm and did not survive.

One focus of this research effort is on firms that started as nonemployer firms but transitioned to becoming employer firms. Those firms can be found in bucket two in the set of four buckets and buckets two and five in the set of six buckets. A second focus is on firm survival, which are firms in the first four buckets of the set of six. The final focus of this research is to examine positive net job creation over the period. Net job creation over the period comes from firms in buckets two and four in the second set of six buckets.

In summary, this research seeks to examine what traits delineate firms that:

1) transition from nonemployer to employer versus those that remain nonemployers;

2) survive over the period, versus those that close; and

3) have positive net job creation over the period.

Using the full CSP, we first examine the traits of the 2011/2012 startups in each bucket and make some initial comparisons. We then examine the subsample of those startups from the CSP that were matched to the SBO. Finally, we explore the transitions, survival, and job creation in a multivariate setting.

Previous work using the CSP by Fairlie et al (2023) showed that there were about 4.1 million businesses that are started each year in the United States and nearly 90 percent of them were nonemployer startups. Our research uses the universe of startups that began operations in 2011 or 2012. As shown in Table 1, data from the Comprehensive Startup Panel indicate that there were more than 4.3 million businesses that started in each of the years 2011 and 2012.

Table 1. Startup Businesses

Startup Cohort	Number of Firms	Percentage
2011 Start	4,387,000	50.1%
2012 Start	4,365,000	49.9%
Total	8,752,000	100%

Source: Comprehensive Startup Panel, 2011-2012 Startup Cohorts.

For our analysis, the 2011 startups were tracked from 2011-2018 and the 2012 startups were tracked over the 2012-2019 period. Overall, the vast majority of businesses, about 89 percent, started as a nonemployer and stayed as a nonemployer over the eight years of observation. As shown in Table 2, only 1.8 percent of firms started as a nonemployer and transitioned to an employer firm. Just over nine percent of businesses started as employer businesses. About two-thirds of those did not have any net job creation over the period and about one-third had positive net job creation over the period.

Table 2. Firm Transition Status

Status	Number of Firms	Percentage
Started and ended as a non employer	7,801,000	89.1%
Started as a nonemployer but transitioned to employer during period	159,400	1.8%
Started as an employer; had no net job creation over period	522,500	6.0%
Started as an employer; had positive net job creation over period	268,700	3.1%
Total	8,752,000	100%

Source: Comprehensive Startup Panel, 2011-2012 Startup Cohorts.

For the two percent of businesses that started as nonemployers and became employer firms over the eight year period of observation, it's interesting to note that the transition happened in the first year of observation for more than five percent of firms (see Table 3). It's worth noting that we can't definitively know that a firm was founded as a nonemployer if it was observed to have employees within the first year. It could be that these firms really do reflect meaningful nonemployer-to-employer transitions, however they could be employer firms from the first day they "opened their doors" that were mischaracterized as nonemployer startups

because of the natural pace of building up the business. While this is a valid concern, we make the assumption that they were true non-employer-to-employer transitions. Given the largest groups are the ones that transition in years 1 and 2 (rather than year 0) it's reasonable to assume that this issue is not driving results even if they were misclassified.

More than three-quarters of the firms that transitioned did so in the first four years of operation. Beginning in the fifth year of operation, the share of firms that transitioned fell to less than eight percent and decreased even further in the three subsequent years. This seems to indicate that if firms are going to transition from nonemployer to employer, they are more likely to do so in the early years of their operations, which is consistent with the findings of Fairlie and Miranda (2017) and Fairlie et al. (2024).

Table 3. Year of Transition

Year	Number of Firms	Percentage	Cumulative Percentage
0	8,867	5.6%	5.6%
1	70,470	44.2%	49.8%
2	27,170	17.0%	66.8%
3	16,770	10.5%	77.3%
4	12,070	7.6%	84.9%
5	8,897	5.6%	90.5%
6	8,482	5.3%	95.8%
7	6,703	4.2%	100%
Total	159,400	100%	

Source: Comprehensive Startup Panel, 2011-2012 Startup Cohorts.

We now expand the number of buckets from four to six to take into account survival over the period. These results are found in Table 4. Less than a quarter of businesses started and stayed as nonemployers and survived over the period. Just over one percent of businesses (1.3%) started as nonemployers and transitioned to becoming employer firms and survived over the period. About two-thirds of businesses (66.4 percent) started as nonemployers and closed down over the period. Only about 1.3 percent of businesses started as employer firms and

survived but had no net new job creation. Just over three percent of businesses started as employers, survived, and had net new job creation over the period. Just under five percent of businesses (4.6 percent) started as employers but did not survive over the period.

Table 4. Firm Survival Status

Status	Number of Firms	Percentage
Started and ended as a nonemployer; survived	2,036,000	23.3%
Started as a nonemployer but transitioned to employer during period; survived	109,400	1.3%
Started as an employer; had no net job creation over period; survived	117,000	1.3%
Started as an employer; had positive net job creation over period; survived	268,700	3.1%
Started as a nonemployer and didn't survive	5,815,000	66.4%
Started as an employer and didn't survive	405,400	4.6%
Total	8,752,000	100%

Source: Comprehensive Startup Panel, 2011-2012 Startup Cohorts.

Turning to survival rates, which can be found in Table 5, overall less than 30 percent of businesses survived over the eight year period of observation. For businesses that started and ended the period as nonemployers, the survival rate was just 26.1 percent. Interestingly, businesses that started as employers but had no net job creation over the period actually had the lowest survival rate at 22.4 percent. Businesses that started as nonemployers but transitioned to become employer firms over the period had a survival rate of 68.6 percent. Overall, nonemployer startups had a survival rate of 27 percent, while employer startups had a survival rate of nearly 49 percent.

Table 5. CSP Survival Rate for 2011 and 2012 Startups

Status	Survival Rate
Started and ended as a non employer	26.1%
Started as a nonemployer but transitioned to employer during period	68.6%
Started as an employer; had no net job creation over period	22.4%
Started as an employer; had positive net job creation over period	100%
Total	28.9%
Started as an Employer	48.8%
Started as a Nonemployer	27.0%
Total	28.9%
Base Year Employment	
0	28.2%
1	44.5%
2-4	51.2%
5-9	54.1%
10-19	52.8%
20-49	53.4%
50-99	57.4%
100 or more	57.9%
Total	28.9%

Source: Comprehensive Startup Panel, 2011-2012 Startup Cohorts.

Survival rates for startups differed dramatically by firm employment size at startup. For example, firms with no employees had a survival rate of just 28.2 percent, while businesses that started with one employee had a survival rate of 44.5 percent. For businesses with more than one employee, the survival rate increased as the initial year employment increased.

However, as shown in Table 6, a little over 3 percent of the 2011-2012 startups had employees and the vast majority of these employer startups had less than five employees.

Table 6. Employment size

Number of Employees	Number of Firms	Percentage
0	8,469,000	96.8%
1	95,270	1.1%
2-4	101,800	1.2%
5-9	43,220	0.5%
10-19	22,100	0.3%
20-49	12,210	0.1%
50-99	4,108	0.0%
100 or more	3,662	0.0%
Total	8,752,000	100.0%

Source: Comprehensive Startup Panel, 2011-2012 Startup Cohorts.

A few things to note about these data before we move to the matched sample. Recall that startups are defined as starting up in the first year when they have positive revenues. Employer startups are technically those businesses that have positive payroll or any paid employees in their first year of operation. Employment is measured as the number of employees on the payroll in the pay period that included March 12th. Therefore, there are a number of firms with zero employment in that specific pay period but do have annual payroll; these could be seasonal businesses, businesses that started after March 12th, or that hired their first employee after the March 12th pay period. These are still considered employer startups because they had positive annual payroll in their first year of having revenues.

A nuanced look at nonemployers in the data may be helpful when examining the full 2011-2012 CSP to the matched sample discussed below. Recall that the CSP includes every business activity that files taxes, which includes a large number of business activities that include consulting, contracting, and hobby activities. Many of these activities are represented by nonemployers, who make up the vast majority of startups, that have no intention of hiring employees or growing a business. One question is whether or not these are actual businesses or just business

activities. For example, a professor who gives a couple of lectures and receives honorariums for those lectures would require him or her to file this revenue with the Internal Revenue Service (IRS). These kinds of activities would typically be found in a Form 1040 Schedule C filing on an individual's tax return and filed with the IRS through the person's social security number.

Fairlie and his colleagues (2023) investigate an alternative definition of startup that excludes secondary jobs and smaller-scale self-employment activities by narrowing their definition of nonemployer startups to exclude sole proprietorships that do not have an employer identification number (EIN); the EIN is used by businesses to fulfill their tax obligations to the IRS and many businesses have EINs even when they don't have employees. When we look at the full cohort of 2011 and 2012 startups, 2.3 million out of 8.75 million of these businesses started with an EIN and about two-thirds of those were nonemployers. Of the nearly 160,000 nonemployers that transitioned, only about 25,000 of them started without an EIN. When we look at the transition rate overall (two percent), it was much lower than the transition rate of those nonemployers that started with an EIN (8.9 percent).

Matched CSP - SBO Sample

The next set of tables summarizes the data from the subsample of data from the Comprehensive Startup Panel data that were matched to the 2012 Survey of Business Owners. This sample contains just over 100,000 businesses. The following tables make use of the survey weights provided in the SBO, which take into account the survey methodology of stratification and sampling, to generate estimates of the underlying business population that the sampled respondents represent. Thus, the weighted total for the sample of 100,000 businesses is on the order of 1.3 million businesses.

The share of nonemployers of this matched subset of startups is around 84 percent, while the share of employers is around 16 percent. As shown in Table 7, our analysis using the four buckets for firm status shows us about 82 percent of startups were nonemployers that remained nonemployers over the period of observation. About 2.5 percent of startups were nonemployers that transitioned to employers over the period. Nine percent of businesses were employers that had no positive net job creation over the period, while 6.7 percent were employer startups that experienced positive net job creation.

When we expand the analysis to six buckets in order to take survival into account, we see that 29.5 percent of startups were nonemployer startups that remained nonemployers and survived over the period, while 1.8 percent were nonemployer startups that transitioned to employers and survived. More than 53 percent of startups were nonemployer startups that did not survive over the period. Just over 6 percent were employer startups that did not survive. Nearly three percent of startups were employer startups that survived but had no net positive job creation over the period and 6.7 percent were employer startups that had positive net job creation over the period and survived.

Table 7. Firm Status for CSP-SBO Matched Sample (Weighted)

Transition Status	**Number of Firms**	**Percentage**
Started and ended as a non employer	1,072,000	81.9%
Started as a nonemployer but transitioned to employer during period	32,010	2.5%
Started as an employer; had no net job creation over period	117,200	9.0%
Started as an employer; had positive net job creation over period	87,230	6.7%
Total	1,309,000	100%
Survival Status	**Number of Firms**	**Percentage**
Started and ended as a nonemployer; survived	385,800	29.5%
Started as a nonemployer but transitioned to employer during period; survive	23,060	1.8%
Started as an employer; had no net job creation over period; survived	37,390	2.9%
Started as an employer; had positive net job creation over period; survived	87,230	6.7%
Started as a nonemployer and didn't survive	695,400	53.1%
Started as an employer and didn't survive	79,800	6.1%
Total	1,309,000	100%

Source: Matched 2011-2012 Sample of Startups from Comprehensive Startup Panel and 2012 Survey of Business Owners.

As shown in Table 8, employer startups that did not have positive net job creation over the period were the group with the lowest survival rate overall (31.9 percent). Nonemployer startups that remained nonemployers had the next lowest survival rate at 36 percent. Nonemployer startups that transitioned to employers had a survival rate of 72 percent. Overall, the survival rate for the whole sample was 40.8 percent, which is higher than the survival rate of the full CSP cohort at 28.9 percent.

Table 8. Firm Status and Survival Rates: CSP-SBO Matched Sample (Weighted)

Status	**Survival Rate**
Started and ended as a non employer	36.0%
Started as a nonemployer but transitioned to employer during period	72.1%
Started as an employer; had no net job creation over period	31.9%
Started as an employer; had positive net job creation over period	100.0%
Total	40.8%

Source: Matched 2011-2012 Sample of Startups from Comprehensive Startup Panel and 2012 Survey of Business Owners.

Breaking out the matched sample by firm status and legal form of organization, we see in Table 9 that about two-thirds of corporations were employer startups, versus only a quarter of partnerships and less than three percent of sole proprietorships (and a small number of other legal forms like nonprofit organizations, which were included with sole proprietorships). Corporations had the highest share of startups that were nonemployer that transitioned to employer at more than five percent. Just over three percent of partnerships started as nonemployers but transitioned to employers. Less than one percent of sole proprietors (0.8%) started as nonemployers and transitioned to employers. In terms of survival, corporations were the most likely to survive over the period (57.5 percent), while sole proprietors had the lowest survival rate (34.6 percent).

Table 9. Firm Status and Legal Form of Organization: CSP-SBO Matched Sample (Weighted)

Status	Corporation	Partnership	Sole Prop/Other	Total
Started and ended as a nonemployer; survived	10.1%	37.9%	32.5%	29.5%
Started as a nonemployer but transitioned to employer during period; survived	5.1%	3.2%	0.8%	1.8%
Started as an employer; had no net job creation over period; survived	12.9%	3.8%	0.4%	2.9%
Started as an employer; had positive net job creation over period; survived	29.4%	10.2%	0.9%	6.7%
Started as a nonemployer and didn't survive	18.8%	33.9%	64.1%	53.1%
Started as an employer and didn't survive	23.6%	11.1%	1.3%	6.1%
Total	100.0%	100.0%	100.0%	100.0%
Started as an employer	65.9%	25.1%	2.6%	15.6%
Started as a nonemployer	34.1%	74.9%	97.4%	84.4%
Survived	57.5%	55.1%	34.6%	40.8%
Did not Survive	42.5%	44.9%	65.4%	59.2%
Source: Matched 2011-2012 Sample of Startups from Comprehensive Startup Panel and 2012 Survey of Business Owners.				

The amount of financial capital used in the year of starting up is correlated with the kind of startup (employer versus nonemployer) as well as outcome measures over time. As shown in Table 10, as the amount of financial capital investment increases, so does the share of employer startups going from a low of 5.7 percent of firms with no financial capital investment to a high of 46.5 percent for firms with $250K to $1 Million. This relationship holds up until the highest level of financial capital investment where we see 37 percent of firms that started with more than

a million dollars of financial capital were employer startups. In terms of survival, higher levels of financial capital are associated with higher survival rates. The survival rate also increases with financial capital investment, ranging from a low of 32.4 percent for businesses no financial capital investment to a high of 67% for startups with more than one million dollars of investment. In terms of transition from nonemployer to employer, nonemployer startups with higher levels of financial capital investment tended to have a higher share that transitioned to employer firms than those with the lowest levels of financial capital investment.

Table 10. Firm Status by Startup Financial Capital: CSP-SBO Matched Sample (Weighted)

Status	$0	<$5K	$5K-$10K	$10K-$25K	$25K-$50K	$50K-$100K	$100K-$250K	$250K-$1M	>$1M	Total
Started and ended as a nonemployer; survived	28.2%	33.3%	31.4%	27.5%	24.8%	23.3%	26.0%	31.1%	40.3%	29.8%
Started as a nonemployer but transitioned to employer during period; survived	0.7%	1.5%	3.6%	3.7%	4.3%	4.2%	2.9%	2.1%	2.3%	1.8%
Started as an employer; had no net job creation over period; survived	1.2%	2.1%	4.4%	5.5%	5.6%	6.7%	6.7%	7.2%	6.3%	2.8%
Started as an employer; had positive net job creation over period; survived	2.2%	3.7%	8.7%	12.1%	15.4%	20.8%	21.1%	24.0%	18.1%	6.6%
Started as a nonemployer and didn't survive	65.3%	55.5%	44.4%	39.3%	35.7%	28.4%	26.9%	20.3%	20.5%	53.1%
Started as an employer and didn't survive	2.2%	4.0%	7.6%	11.9%	14.2%	16.6%	16.5%	15.4%	12.5%	5.9%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Started as an employer	5.7%	9.8%	20.7%	29.5%	35.2%	44.1%	44.3%	46.5%	37.0%	15.3%
Started as a nonemployer	94.3%	90.2%	79.3%	70.5%	64.8%	55.9%	55.7%	53.5%	63.0%	84.7%
Survived	32.4%	40.6%	48.0%	48.7%	50.1%	55.0%	56.7%	64.3%	67.0%	41.0%
Did not Survive	67.6%	59.4%	52.0%	51.3%	49.9%	45.0%	43.3%	35.7%	33.0%	59.0%

Source: Matched 2011-2012 Sample of Startups from Comprehensive Startup Panel and 2012 Survey of Business Owners.

In terms of owner demographics, the SBO defines firms by owner sex, race, ethnicity, and veteran status at the firm level. For example, if a firm was 60% owned by women, it would be women-owned. If it was 50% owned by women and 50% owned by men it would be defined as equally-owned. For our analysis, we define a firm to be women-owned, minority-owned, etc. if the firm is owned by 50 percent or more of that group. The next table breaks out business status by owner sex, minority status, and veteran status. Minority-owned indicates that the firm is owned by owners who are non-white and/or Hispanic.

As shown in Table 11, men were slightly more likely to start businesses as employers and women were more likely to start as nonemployers. Non-minority-owned firms were more likely

to be employer startups than minority-owned firms. Non-Hispanic-owned firms were more likely to be employer startups than Hispanic-owned firms. Veterans were less likely to be employer startups than non-veterans.

In terms of survival, male-owned businesses had a higher rate of survival than female-owned businesses; minority-owned firms had a slightly lower rate of survival than non-minority-owned businesses. Hispanic-owned businesses had a lower survival rate than non-Hispanic-owned businesses and veteran-owned businesses had a lower survival rate than non-veteran-owned businesses. Around 30 percent of firms, regardless of owner demographics, started and ended as nonemployers and survived.

Table 11: Firm Status by Owner Demographics: CSP-SBO Matched Sample (Weighted)

Status	Male	Female	Minority	Non-Minority	Hispanic	Non-Hispanic	Veteran	Non-Veteran	Total
Started and ended as a nonemployer; survived	29.6%	30.3%	28.6%	29.8%	29.3%	29.5%	27.1%	29.7%	29.5%
Started as a nonemployer but transitioned to employer during period; survived	2.2%	1.5%	1.8%	1.7%	1.9%	1.7%	1.5%	1.8%	1.8%
Started as an employer; had no net job creation over period; survived	3.4%	2.4%	2.5%	3.0%	2.0%	3.0%	2.5%	2.9%	2.9%
Started as an employer; had positive net job creation over period; survived	8.0%	5.3%	6.1%	6.9%	4.9%	6.9%	5.2%	6.8%	6.7%
Started as a nonemployer and didn't survive	50.0%	55.5%	54.9%	52.5%	57.9%	52.5%	57.6%	52.7%	53.1%
Started as an employer and didn't survive	6.9%	5.1%	6.1%	6.1%	3.9%	6.4%	6.2%	6.1%	6.1%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%
Started as an employer	18.2%	12.7%	14.6%	16.0%	10.8%	16.2%	13.8%	15.8%	15.6%
Started as a nonemployer	81.8%	87.3%	85.4%	84.0%	89.2%	83.8%	86.2%	84.2%	84.4%
Survived	43.1%	39.4%	39.0%	41.4%	38.1%	41.1%	36.2%	41.2%	40.8%
Did not Survive	63.5%	64.6%	65.3%	64.1%	66.7%	64.1%	66.7%	64.2%	64.4%

Source: Matched 2011-2012 Sample of Startups from Comprehensive Startup Panel and 2012 Survey of Business Owners.

Firm status by race can be found in Table 12. Black-owned and other (Native Hawaiian and Other Pacific Islander (NHOPI) and American Indian and Alaska Native (AIAN)) had the lowest shares of employer startups. Businesses owned by Asians and companies that were not classifiable by race/gender/ethnicity had the highest shares of employer startups. In terms of survival, businesses owned by blacks, Native Hawaiian and Other Pacific Islander (NHOPI), and American Indian and Alaska Native (AIAN) had the lowest survival rates, while businesses

owned by whites and Asians had the highest. In terms of transition, black-owned businesses, and companies that were not classifiable by race/gender/ethnicity had the lowest share of nonemployer startups that transitioned to employers. Asian-owned and companies that were not classifiable by race/gender/ethnicity had businesses that had the highest share of employer startups that survived and had positive net job growth over the period.

Table 12. Firm Status by Owner Demographics: CSP-SBO Matched Sample (Weighted)

Status	White	Black	Asian	Other Race	Public/ Not Classifiable	Total
Started and ended as a nonemployer; survived	30.0%	29.4%	27.5%	28.5%	25.6%	29.5%
Started as a nonemployer but transitioned to employer during period; survived	1.8%	1.1%	1.9%	1.7%	0.6%	1.8%
Started as an employer; had no net job creation over period; survived	2.9%	1.5%	3.6%	1.7%	3.3%	2.9%
Started as an employer; had positive net job creation over period; survived	6.6%	3.1%	9.3%	4.1%	9.2%	6.7%
Started as a nonemployer and didn't survive	53.2%	60.9%	48.1%	60.4%	47.9%	53.1%
Started as an employer and didn't survive	5.7%	4.0%	9.6%	3.6%	13.5%	6.1%
Total	100%	100%	100%	100%	100%	100%
Started as an employer	15.1%	8.6%	22.5%	9.4%	26.0%	15.6%
Started as a nonemployer	84.9%	91.4%	77.5%	90.6%	74.0%	84.4%
Survived	41.2%	35.1%	42.4%	36.0%	38.7%	40.8%
Did not Survive	58.8%	64.9%	57.6%	64.0%	61.3%	59.2%
Source: Matched 2011-2012 Sample of Startups from Comprehensive Startup Panel and 2012 Survey of Business Owners.						

Multivariate Analyses

We now examine startup dynamics in a multivariate setting. The intent of the Foundations for Evidence-Based Policymaking Act of 2019 is "to improve the use of evidence and data to generate policies and inform programs" supported by government (Fairlie, et al. (2023) p. 178). This too is an objective motivating our research. The newly available Comprehensive Startup Panel (CSP) assembled by Robert Fairlie and his Census co-authors is a resource with many applications in entrepreneurship research. These data and their findings are not an end product but rather important catalysts encouraging more research on entrepreneurial dynamics. Quoting Fairlie et al. (2023), p. 187, "Measuring job creation and survival rates using a broad definition versus a narrow definition of entrepreneurs is relevant for policy discussions."

Their broad entrepreneurship definition entailed equating firm ownership with filing business income tax returns. We employ this definition and use the entire matched sample in our initial multivariate analyses, but we repeat our analyses using a narrower definition to identify a subset of "substantive" startups as will be defined in more detail below. We also provide separate analyses for women-owned firms, minority-owned firms, and businesses in human-capital-intensive industries in order to investigate whether there are factors that affect some kinds of firms' outcomes differently.

The three outcomes for our multivariate analysis of startup dynamics are:

1) the transition of nonemployer startups from 2011/2012 to becoming employer firms over the seven year period following startup;

2) survival over the seven year period after startup; and

3) positive net employment growth over the period. This would include nonemployer startups that transitioned to employer businesses and employer startups that added new employees (over their original base year employment) over the period of observation.

Transition from Nonemployer to Employer

Our regression models employ standard logistic regressions. The first regression model examines the transition from nonemployer to employer status over seven years following startup. This model includes over 67,000 businesses that began as nonemployers. The dependent variable is binary, taking the value of one if the business becomes an employer and zero if it remains a nonemployer.

The independent variables encompass a series of binary and categorical variables describing various characteristics of the business and its owner. Examples of binary variables include

whether the business is a franchise, home-based, the primary source of income for the owner, or if the owner had prior business experience. Categorical variables include the legal form of the organization: sole proprietorship, partnership, corporation, or other legal forms like tax-exempt organizations. In the regression model, one category is excluded as the reference group, and the remaining categories are interpreted relative to the excluded category.

The statistical significance of a coefficient in logistic regressions is assessed using the z-statistic, which is also known as the Wald statistic. The three common threshold levels for statistical significance are associated with different p-values: 1) 0.05 level (5%): This is the most common threshold. A z-statistic is considered statistically significant at this level if the corresponding p-value is less than 0.05. This typically corresponds to a z-statistic of approximately ±1.96. If the z-statistic is beyond ±1.96, the coefficient is considered statistically significant at the 5% level; 2) 0.01 level (1%): This more stringent threshold is used when a stronger level of evidence is required. The z-statistic is considered statistically significant at this level if the p-value is less than 0.01. This corresponds to a z-statistic of approximately ±2.58. If the z-statistic is beyond ±2.58, the coefficient is considered statistically significant at the 1% level; 3) 0.001 level (0.1%): This is an even more stringent level of significance, often used in very large datasets or where very strong evidence is needed. The z-statistic is considered statistically significant at this level if the p-value is less than 0.001. This corresponds to a z-statistic of approximately ±3.29. If the z-statistic is beyond ±3.29, the coefficient is considered statistically significant at the 0.1% level. These thresholds are standard in hypothesis testing, indicating the probability of observing a result as extreme as, or more extreme than, the observed one if the null hypothesis were true. The smaller the p-value, the stronger the evidence against the null hypothesis. When we refer to a

result being statistically significant, we are saying it meets at least the minimum 5% significance level.

To illustrate, let's examine the result where the dependent variable is the transition from a nonemployer to an employer firm, and the predictor variable is the legal form of the business (corporation vs. sole proprietorship). A coefficient of 2.75 indicates that the log-odds of transitioning from a nonemployer to an employer firm are 2.75 units higher for corporations compared with sole proprietorships holding all other variables constant. To convert this to an odds ratio, we exponentiate the coefficient. The odds ratio is $e^{2.75} \approx 15.61$, which means corporations have approximately 15.61 times the odds of making this transition compared with sole proprietorships, holding all other factors constant. In simpler terms, being a corporation (compared to a sole proprietorship) is associated with a significantly higher likelihood of the outcome occurring. The coefficient is highly significant (well above the .01 significance level). The coefficient on partnership is also positive and highly statistically significant, indicating they are also more likely than sole proprietorships to transition to employer status. These findings provide strong evidence that the legal form is related to the likelihood of transitioning to an employer firm. This finding aligns with the literature, which suggests that legal structure can influence business outcomes (Djankov et al., 2002; Klapper, Laeven, & Rajan, 2006; Novik and Luque, forthcoming).

Table 13. Logistic Regression on Transition from Nonemployer to Employer Firm

	Coefficient	Std. err.	Z Statistic
Corporation	2.7500	0.0500	55.00
Partnership	1.8920	0.0675	28.03
< $5K FK investment	0.3130	0.0598	5.24
$5K-$10K FK investment	0.7334	0.0743	9.88
$10-$25K FK investment	0.7301	0.0733	9.96
$25-$50K FK investment	0.8217	0.0838	9.80
$50K-$100K FK investment	0.8942	0.0880	10.16

$100K-$250K FK investment	0.8394	0.0935	8.98
$250K-$1M FK investment	0.7267	0.1219	5.96
$1M+ FK investment	0.4859	0.1872	2.60
Team Ownership	-0.2871	0.0574	-5.00
Joint Ownership- Spouses	0.1682	0.0661	2.54
Intellectual Property	0.1071	0.0727	1.47
Has Business and/or Government Clients	0.1254	0.0440	2.85
Home based	-0.4905	0.0444	-11.06
Exporter	-0.1208	0.1006	-1.20
E-Commerce	-0.1951	0.0636	-3.07
Website	0.6108	0.0458	13.34
Construction	0.5880	0.1713	3.43
Manufacturing	0.2796	0.1727	1.62
Wholesale	0.2863	0.1816	1.58
Retail	0.5010	0.1642	3.05
Transportation & Warehousing	0.0996	0.1771	0.56
Information, Finance, & Insurance	0.1549	0.1691	0.92
Real Estate	-0.5984	0.1798	-3.33
Professional Services	0.3180	0.1625	1.96
Other Services	0.2579	0.1718	1.50
Administration & Support Services	0.4943	0.1762	2.80
Education Services	0.0657	0.2154	0.31
Health Care and Social Assistance	0.3990	0.1779	2.24
Arts, Entertainment, and Recreation	0.0016	0.1874	0.01
Accommodation and Food Services	0.7228	0.1852	3.90
Female	-0.0444	0.0459	-0.97
Black	-0.3913	0.0982	-3.98
Asian	-0.0385	0.0688	-0.56
Native Hawaiian & Other Pacific Islander	D	D	D
American Indian & Alaskan Native	-0.0381	0.1813	-0.21
Other Race	0.0158	0.1286	0.12
Hispanic	-0.0665	0.0725	-0.92
Veteran	-0.1191	0.0770	-1.55
Not Classifiable	0.4320	0.1693	2.55
Franchise	-0.0774	0.1027	-0.75
Owner Age 25-34	0.4259	0.1013	4.20
Owner Age 35-44	0.2907	0.1013	2.87
Owner Age 45-54	0.1322	0.1025	1.29
Owner Age 55-64	-0.0657	0.1087	-0.60
Owner Age: 65 plus	-0.0647	0.1324	-0.49
Owner Education- Some College	-0.0675	0.0591	-1.14
Owner Education- College Degree	0.0213	0.0606	0.35
Owner Education- Graduate Degree	0.0878	0.0692	1.27
Previous Business Experience	0.0979	0.0446	2.20
Primary Source of Income	0.2546	0.0490	5.20
Hours Worked 20-39	0.5570	0.0653	8.53

Hours Worked 40	0.7397	0.0761	9.72
Hours Worked 41-59	1.0370	0.0675	15.36
Hours Worked 60 plus	1.1520	0.0689	16.71
Founded	-0.2174	0.0634	-3.43
Constant	-5.1520	0.1888	-27.29

N (unweighted) = 67,500
R2 = 0.2838
Note: "D" means suppressed to ensure that no confidential information is disclosed.

Financial capital investment is another critical factor influencing the likelihood of transition. The excluded or reference category in the model is using no financial capital investment and eight levels of investment are the included categories. Coefficients for all categories are positive and statistically significant, which provides strong evidence of the importance of adequate financial capital in supporting a firm's transition from nonemployer to employer. This finding is consistent with theories of capital constraints and entrepreneurial growth (Evans & Jovanovic, 1989; Hurst & Lusardi, 2004).

Additional firm characteristics also play a significant role. Team ownership (more than one owner), being home-based, and having e-commerce capabilities are negatively associated with the likelihood of transition. Conversely, joint ownership by spouses, having a website, and having business and/or government clients positively influence the transition. These results suggest that a business's operational setup and customer base are crucial determinants of growth potential. The positive association with having a website is consistent with a study from Bertschek et al. (2013), which highlights the importance of digital presence for business expansion. Industry-specific effects are also evident; construction, retail, administration and support services, health care and social assistance, and accommodation and food services are positively associated with transition, while real estate is negatively associated. This differentiation in industry impact

is supported by Audretsch (1995), who found that industry characteristics significantly influence business dynamics and growth trajectories.

Owner demographics, such as sex, race, and ethnicity, generally do not significantly influence the transition. However, being black is negatively associated with the likelihood of transitioning, while companies not classifiable by race/gender/ethnicity are more likely to transition. Business owners ages 25 to 44 are more likely to transition compared with businesses with the youngest owners (under 25), however the coefficients on the oldest age categories were not statistically significant. This finding is consistent with empirical evidence showing that younger entrepreneurs are often more dynamic and willing to take risks (Lévesque & Minniti, 2006). Education levels surprisingly did not appear to be correlated with the likelihood of transitioning, however previous business experience had a positive and statistically significant effect, as did having a business that provides the primary source of income. Hours worked per week is a significant positive factor; more hours worked correlates with a higher likelihood of transitioning, emphasizing the importance of effort and dedication in business success. Finally, being founded by the owner (as opposed to inherited, bought, or gifted) was negatively association with transitioning from nonemployer to employer.

Some firm-level characteristics did not appear to have any statistically significant effects on the transition. These include having any intellectual property (e.g., copyrights, trademarks, patents), franchise, and exporting. These findings suggest that while these factors may contribute to overall business strategy, they do not significantly affect the transition from nonemployer to employer status within this dataset.

Subsample Analysis and Special Groups

When the logistic regression is rerun on a subset of substantive firms (those with some financial capital investment, founded by the owner, and where the owner works at least 20 hours per week), the results remain broadly consistent. In this model, the excluded categories in two cases changed: Less than $5K in financial capital investment and 20-39 hours worked. The other control variables remained the same except for founded, which by definition is dropped because the subsample contained only businesses that were founded by the owners. This decreased the number of observations from more than 67,000 to around 22,500. The results are found in Table 14.

All reported financial capital investment levels were positive and statistically significant at least at the 1% level except for the category of more than one million dollars. Joint ownership by spouses, E-commerce, and previous business experience were no longer statistically significant. Only the 25-34 age category was statistically significant. Given these findings, when we run the regressions separately for women-owned firms, minority-owned firms, and human-capital-intensive industries, we'll use the full sample to ensure sufficient sample sizes for those regressions.

Table 14. Logistic Regression on Transition from Nonemployer to Employer Firm: Substantive Firms

	Coefficient	Std. err.	Z Statistic
Corporation	2.5340	0.0616	41.16
Partnership	1.8070	0.0843	21.44
$5K-$10K FK investment	0.5250	0.0763	6.88
$10-$25K FK investment	0.4168	0.0777	5.36
$25-$50K FK investment	0.4923	0.0936	5.26
$50K-$100K FK investment	0.6522	0.1007	6.47

$100K-$250K FK investment	0.5448	0.1146	4.75
$250K-$1M FK investment	0.5277	0.1552	3.4
$1M+ FK investment	D	D	
Team Ownership	-0.3505	0.0730	-4.8
Joint Ownership- Spouses	0.1266	0.0835	1.52
Intellectual Property	-0.0458	0.0887	-0.52
Has Business and/or Government Clients	0.1706	0.0562	3.04
Home based	-0.4750	0.0548	-8.66
Exporter	0.0206	0.1184	0.17
E-Commerce	-0.1436	0.0770	-1.87
Website	0.5984	0.0574	10.42
Construction	0.5596	0.2201	2.54
Manufacturing	0.2462	0.2232	1.1
Wholesale	0.1401	0.2373	0.59
Retail	0.4036	0.2137	1.89
Transportation & Warehousing	0.0895	0.2264	0.4
Information, Finance, & Insurance	0.1493	0.2207	0.68
Real Estate	-0.5379	0.2446	-2.2
Professional Services	0.2670	0.2111	1.26
Other Services	0.1154	0.2241	0.52
Administration & Support Services	0.4860	0.2292	2.12
Education Services	0.1651	0.2787	0.59
Health Care and Social Assistance	0.4551	0.2296	1.98
Arts, Entertainment, and Recreation	-0.0512	0.2425	-0.21
Accommodation and Food Services	0.7210	0.2426	2.97
Female	0.0510	0.0584	0.87
Black	-0.3878	0.1172	-3.31
Asian	-0.0338	0.0898	-0.38
Native Hawaiian & Other Pacific Islander	D	D	D
American Indian & Alaskan Native	0.2681	0.2092	1.28
Other Race	0.0516	0.1544	0.33
Hispanic	-0.0294	0.0880	-0.33
Veteran	-0.0151	0.0958	-0.16
Franchise	-0.2060	0.1425	-1.45
Owner Age 25-34	0.3870	0.1434	2.7
Owner Age 35-44	0.2465	0.1436	1.72
Owner Age 45-54	0.0380	0.1455	0.26
Owner Age 55-64	-0.1550	0.1538	-1.01
Owner Age: 65 plus	-0.1215	0.1875	-0.65
Owner Education- Some College	-0.1271	0.0747	-1.7
Owner Education- College Degree	-0.0708	0.0776	-0.91

Owner Education- Graduate Degree	0.0371	0.0887	0.42
Previous Business Experience	0.0478	0.0553	0.86
Primary Source of Income	0.1967	0.0588	3.35
Hours Worked 40	0.1907	0.0848	2.25
Hours Worked 41-59	0.4816	0.0731	6.59
Hours Worked 60 plus	0.6108	0.0748	8.17
Constant	-4.2500	0.2530	-16.8

N (unweighted) = 22,500
R2 = 0.2312
Note: "D" means suppressed to ensure that no confidential information is disclosed.

The results for women-owned businesses can be found in Table 15. There were more than 32,000 businesses that were women-owned in our sample. In terms of firm characteristics, one difference from the full sample was that the coefficient on having business and/or government clients was not statistically significant. The only industry dummies that were statistically significant were construction, professional services, health care and support services, and accommodation and food services. In terms of owner characteristics, the main differences were the coefficients on black and founder were no longer statistically significant and the coefficient on the highest education level, graduate degree, was positive and statistically significant.

Table 15. Logistic Regression on Transition from Nonemployer to Employer Firm: Female-Owned Firms

	Coefficient	Std. err.	Z Statistic
Corporation	2.8660	0.0826	34.69
Partnership	1.7950	0.1072	16.73
< $5K FK investment	0.3565	0.0958	3.72
$5K-$10K FK investment	0.8664	0.1199	7.23
$10-$25K FK investment	0.7101	0.1207	5.88
$25-$50K FK investment	1.0550	0.1375	7.67
$50K-$100K FK investment	1.1270	0.1473	7.65
$100K-$250K FK investment	0.8547	0.1630	5.24
$250K-$1M FK investment	0.6913	0.2160	3.20
$1M+ FK investment	D	D	D
Team Ownership	-0.3197	0.1004	-3.19
Joint Ownership- Spouses	0.2340	0.0959	2.44

Intellectual Property	0.0305	0.1271	0.24
Has Business and/or Government Clients	0.0503	0.0742	0.68
Home based	-0.6034	0.0714	-8.45
Exporter	-0.0059	0.1818	-0.03
E-Commerce	-0.2178	0.1046	-2.08
Website	0.5461	0.0742	7.36
Construction	0.8943	0.3806	2.35
Manufacturing	0.6341	0.3625	1.75
Wholesale	0.4105	0.3848	1.07
Retail	0.5177	0.3509	1.48
Transportation & Warehousing	0.7055	0.3743	1.88
Information, Finance, & Insurance	0.5661	0.3619	1.56
Real Estate	-0.3956	0.3734	-1.06
Professional Services	0.7042	0.3492	2.02
Other Services	0.2937	0.3593	0.82
Administration & Support Services	0.7195	0.3682	1.95
Education Services	0.0822	0.4085	0.20
Health Care and Social Assistance	0.7196	0.3581	2.01
Arts, Entertainment, and Recreation	0.3387	0.3791	0.89
Accommodation and Food Services	0.8135	0.3712	2.19
Black	-0.2270	0.1427	-1.59
Asian	-0.0495	0.1089	-0.45
Native Hawaiian & Other Pacific Islander	D	D	D
American Indian & Alaskan Native	D	D	D
Other Race	-0.0813	0.2208	-0.37
Hispanic	-0.0925	0.1110	-0.83
Veteran	-0.1409	0.1517	-0.93
Franchise	-0.0878	0.1723	-0.51
Owner Age 25-34	0.3870	0.1743	2.22
Owner Age 35-44	0.3490	0.1733	2.01
Owner Age 45-54	0.2935	0.1736	1.69
Owner Age 55-64	-0.0300	0.1850	-0.16
Owner Age: 65 plus	-0.0838	0.2279	-0.37
Owner Education- Some College	0.0069	0.0984	0.07
Owner Education- College Degree	0.0233	0.1019	0.23
Owner Education- Graduate Degree	0.2316	0.1119	2.07
Previous Business Experience	0.1557	0.0724	2.15
Primary Source of Income	0.2601	0.0756	3.44
Hours Worked 20-39	0.5649	0.1006	5.62
Hours Worked 40	0.7187	0.1231	5.84
Hours Worked 41-59	1.0960	0.1071	10.23
Hours Worked 60 plus	1.3480	0.1094	12.32
Founded	-0.1725	0.1025	-1.68
Constant	-5.6190	0.3854	-14.58

N (unweighted) = 32,500
R2 = 0.3027

Note: “D” means suppressed to ensure that no confidential information is disclosed.

The results for the 21,000 minority-owned businesses had some interesting differences. The coefficient on team ownership was no longer statistically significant. As with women-owned businesses, the coefficient on having business and/or government clients was not statistically significant. None of the coefficients on industry dummies were statistically significant except the positive coefficient on accommodation and food services. The coefficient on Asian was statistically significant, indicating that Asian-owned businesses are more likely to transition than black-owned businesses. The coefficient on the education category of college degree was negative and statistically significant; the excluded category was high school degree or less. The coefficients on previous business experience and founder were not statistically significant in the model for minority-owned businesses.

Table 16. Logistic Regression on Transition from Nonemployer to Employer Firm: Minority-Owned Firms

	Coefficient	Std. err.	Z Statistic
Corporation	2.8360	0.0933	30.40
Partnership	1.9650	0.1327	14.81
< $5K FK investment	0.5190	0.1176	4.41
$5K-$10K FK investment	0.8557	0.1411	6.06
$10-$25K FK investment	1.0890	0.1363	7.98
$25-$50K FK investment	0.9959	0.1623	6.13
$50K-$100K FK investment	1.2670	0.1681	7.54
$100K-$250K FK investment	0.9776	0.1933	5.06
$250K-$1M FK investment	0.9109	0.2818	3.23
$1M+ FK investment	D	D	D
Team Ownership	-0.0144	0.1092	-0.13
Joint Ownership- Spouses	0.2593	0.1234	2.10
Intellectual Property	-0.0457	0.1541	-0.30
Has Business and/or Government Clients	0.1607	0.0860	1.87
Home based	-0.3054	0.0856	-3.57
Exporter	-0.3269	0.1869	-1.75

E-Commerce	-0.3042	0.1343	-2.26
Website	0.6219	0.0916	6.79
Construction	0.6551	0.3912	1.67
Manufacturing	0.3102	0.3948	0.79
Wholesale	-0.3742	0.4178	-0.90
Retail	0.6530	0.3781	1.73
Transportation & Warehousing	0.0175	0.3973	0.04
Information, Finance, & Insurance	0.2081	0.3926	0.53
Real Estate	-0.6844	0.4190	-1.63
Professional Services	0.1756	0.3799	0.46
Other Services	0.2585	0.3878	0.67
Administration & Support Services	0.1954	0.4041	0.48
Education Services	0.0512	0.4732	0.11
Health Care and Social Assistance	0.5007	0.3934	1.27
Arts, Entertainment, and Recreation	-0.2237	0.4425	-0.51
Accommodation and Food Services	0.8990	0.4031	2.23
Female	-0.0661	0.0852	-0.78
Asian	0.3157	0.1062	2.97
Native Hawaiian & Other Pacific Islander	D	D	D
American Indian & Alaskan Native	0.1995	0.1957	1.02
Other Race	0.1092	0.1330	0.82
Hispanic	0.1854	0.1032	1.80
Veteran	0.2212	0.1617	1.37
Franchise	-0.0442	0.1790	-0.25
Owner Age 25-34	0.4428	0.1895	2.34
Owner Age 35-44	0.3586	0.1887	1.90
Owner Age 45-54	0.1583	0.1942	0.81
Owner Age 55-64	-0.0876	0.2164	-0.40
Owner Age: 65 plus	0.3754	0.2778	1.35
Owner Education- Some College	-0.1986	0.1077	-1.84
Owner Education- College Degree	-0.2674	0.1176	-2.27
Owner Education- Graduate Degree	-0.1222	0.1307	-0.93
Previous Business Experience	0.0259	0.0848	0.31
Primary Source of Income	0.4273	0.0920	4.64
Hours Worked 20-39	0.5032	0.1241	4.06
Hours Worked 40	0.7293	0.1398	5.22
Hours Worked 41-59	0.9262	0.1311	7.07
Hours Worked 60 plus	0.9045	0.1320	6.85
Founded	-0.1612	0.1200	-1.34
Constant	-5.7580	0.4298	-13.40

N (unweighted) = 21,000

R2 = 0.30

Note: “D” means suppressed to ensure that no confidential information is disclosed.

The last group of businesses examined in the transition model was businesses in human-capital-intensive industries, which are industries that are characterized by their reliance on human talent and intellectual capabilities[§§]. Some examples of industries that are human-capital-intensive include Healthcare and Biotechnology, Education and Training, Information Technology (IT) and Software Development, Finance and Consulting, Legal Services, Research and Development (R&D), Creative Industries, and Aerospace and Defense. There were nearly 24,000 businesses in these human-capital-intensive industries in our sample. Running the logistic regression on this subset of businesses yielded some interesting results.

The coefficients on the lowest and three highest dummies for financial capital investment levels were not statistically significant. The coefficients on joint ownership by spouses and business and/or government clients were no longer statistically significant. Interestingly the coefficients on intellectual property, education, and previous business experience were not statistically significant in this subset of high-capital-intensive businesses either. The coefficients on female and franchise were positive and statistically significant. The significance of hours worked remains strong. This highlights the critical role of human capital and intensive labor in driving business growth in these industries (Autor, Levy, & Murnane, 2003).

Table 17. Logistic Regression on Transition from Nonemployer to Employer Firm: Human-Capital-Intensive Industries

	Coefficient	Std. err.	Z Statistic
Corporation	3.1890	0.0875	36.46
Partnership	2.0800	0.1242	16.75
< $5K FK investment	0.1935	0.0983	1.97
$5K-$10K FK investment	0.5451	0.1276	4.27

[§§] See the following sources for more detail on human capital intensity and industry categories: Goldin & Katz (1998); Autor et al (2003); Hecker (2005); OECD (2017).

$10-$25K FK investment	0.5684	0.1327	4.28
$25-$50K FK investment	0.3410	0.1658	2.06
$50K-$100K FK investment	0.6091	0.1750	3.48
$100K-$250K FK investment	0.3350	0.2013	1.66
$250K-$1M FK investment	0.2521	0.2364	1.07
$1M+ FK investment	D	D	D
Team Ownership	-0.2994	0.1049	-2.85
Joint Ownership- Spouses	-0.0046	0.1275	-0.04
Intellectual Property	-0.0041	0.1187	-0.03
Has Business and/or Government Clients	0.0724	0.0745	0.97
Home based	-0.4904	0.0772	-6.35
Exporter	-0.2347	0.2118	-1.11
E-Commerce	-0.0886	0.1149	-0.77
Website	0.8359	0.0789	10.6
Female	0.2406	0.0792	3.04
Black	-0.2995	0.1588	-1.89
Asian	-0.2045	0.1245	-1.64
Native Hawaiian & Other Pacific Islander	D	D	D
American Indian & Alaskan Native	D	D	D
Other Race	-0.2825	0.2734	-1.03
Hispanic	0.0825	0.1269	0.65
Veteran	0.1691	0.1305	1.3
Not Classifiable	0.7243	0.3023	2.4
Franchise	0.4635	0.1983	2.34
Owner Age 25-34	0.5307	0.2074	2.56
Owner Age 35-44	0.4729	0.2071	2.28
Owner Age 45-54	0.2726	0.2098	1.3
Owner Age 55-64	0.1853	0.2179	0.85
Owner Age: 65 plus	0.0542	0.2519	0.21
Owner Education- Some College	-0.2225	0.1465	-1.52
Owner Education- College Degree	0.0511	0.1349	0.38
Owner Education- Graduate Degree	0.2106	0.1349	1.56
Previous Business Experience	0.0568	0.0791	0.72
Primary Source of Income	0.2776	0.0873	3.18
Hours Worked 20-39	0.4937	0.1201	4.11
Hours Worked 40	0.9615	0.1318	7.29
Hours Worked 41-59	1.1490	0.1216	9.45
Hours Worked 60 plus	1.3140	0.1276	10.3
Founded	-0.0882	0.1462	-0.6
Constant	-5.5220	0.2380	-23.2

N (unweighted) = 24,000

R2 = 0.3429

Note: “D” means suppressed to ensure that no confidential information is disclosed.

The transition models highlight several critical insights. The significant impact of legal form on business growth suggests that the initial choice of how to structure the business is an important predicator in the likelihood to transition. The importance of financial capital investment underscores the need for accessible financing options for startups and growing businesses. Programs aimed at reducing capital constraints can help to foster entrepreneurial growth and job creation. The negative impact of being home-based and the positive effect of having a website suggest that digital infrastructure and professional business environments are vital for growth. Support for digital adoption and workspace initiatives can help businesses transition successfully. By integrating these insights into policy frameworks, policymakers can better support entrepreneurial activity and business growth, ultimately fostering a more dynamic and inclusive economy.

Survival Models

The next set of logistic models examines the probability of business survival over the observation period from 2012 to 2019. As seen previously in the univariate tables, the overall survival rate of our matched sample was 40.8%, but that varied by whether or not the firm started with employees in addition to several other factors. Let's delve into other factors, firm characteristics, strategy, and owner characteristics, which might also influence business survival in a multivariate context.

The first set of results, found in Table 18, is for the full sample of 100,000+ businesses. The results show that starting as a nonemployer has a negative effect on the likelihood of survival, even after controlling for a number of firm and owner characteristics. The coefficient on the dummy variable for starting as a nonemployer is -0.4404, indicating that being a

nonemployer (compared to starting as an employer firm) decreases the log-odds of survival by 0.4404. Exponentiating this coefficient gives an odds ratio of approximately 0.643, which means that the odds of survival are approximately 0.643 times the odds of survival for businesses that started as employer firms. In other words, there is about a 35.7% reduction in the odds of survival (since $1-0.643 \approx 0.3571$) for nonemployer startups compared to employer startups.

Businesses that were organized as corporations, partnerships, and other legal forms were all more likely to survive than businesses that were organized as sole proprietorships. Capitalized businesses were more likely to survive than businesses started without financial capital investment. Other firm characteristics that were positively associated with survival include joint ownership between spouses, exporting, E-commerce, and having a website. Industry effects were again mixed, but only a few were statistically significant. Businesses in the information, finance, & insurance industry and real estate sector were more likely to survive, while businesses in accommodation and food services were less likely to survive. Franchises were less likely to survive.

In terms of owner characteristics, businesses owned by women were less likely to survive than businesses owned by men. Businesses owned by blacks and Native Hawaiian and Other Pacific Islanders (NHOPI) were less likely to survive than businesses owned by whites. Veteran-owned businesses were less likely to survive than businesses owned by non-veterans. Businesses that were not classifiable by race/ethnicity/sex were more likely to survive than private businesses and those that were classifiable.

Owner age, education, and hours worked were all important predictors in business survival. The coefficients on all owner age categories were positive and statistically significant

except the one for 65 years and older, which indicates that the youngest owners (under 25 years old) were the least likely to survive. In terms of education, owners with a college degree or graduate degree were more likely to survive than businesses with owners with only a high school education or less. Businesses with owners working more hours than 20 hours a week were more likely to survive than businesses with owners working less than 20 hours a week. Finally, businesses that were the primary source of income for the owner were more likely to survive than businesses that were not the primary source of income.

Table 18. Logistic Regression for Survival: All Firms

	Coefficient	Std. err.	Z Statistic
Started as a Nonemployer	-0.4404	0.0202	-21.78
Corporation	0.4253	0.0219	19.38
Partnership	0.2939	0.0262	11.23
Other Legal Form	1.0350	0.1148	9.01
< $5K FK investment	0.1758	0.0178	9.89
$5K-$10K FK investment	0.2478	0.0279	8.88
$10-$25K FK investment	0.1868	0.0275	6.8
$25-$50K FK investment	0.1632	0.0324	5.03
$50K-$100K FK investment	0.1927	0.0337	5.72
$100K-$250K FK investment	0.2627	0.0344	7.63
$250K-$1M FK investment	0.4018	0.0418	9.62
$1M+ FK investment	0.4920	0.0589	8.35
Team Ownership	-0.0245	0.0203	-1.21
Joint Ownership- Spouses	0.2151	0.0250	8.61
Intellectual Property	-0.0669	0.0273	-2.45
Has Business and/or Government Clients	0.0012	0.0149	0.08
Home based	0.0005	0.0152	0.03
Exporter	0.1125	0.0387	2.9
E-Commerce	0.1144	0.0233	4.92
Website	0.1868	0.0165	11.33
Construction	0.0040	0.0541	0.07
Manufacturing	-0.0028	0.0558	-0.05
Wholesale	-0.0197	0.0606	-0.33
Retail	-0.0652	0.0517	-1.26
Transportation & Warehousing	-0.0319	0.0561	-0.57
Information, Finance, & Insurance	0.1099	0.0538	2.04

Real Estate	0.5299	0.0557	9.51
Professional Services	-0.0302	0.0505	-0.6
Other Services	0.1273	0.0529	2.41
Administration & Support Services	0.0060	0.0543	0.11
Education Services	-0.0197	0.0617	-0.32
Health Care and Social Assistance	0.0108	0.0541	0.2
Arts, Entertainment, and Recreation	0.2332	0.0569	4.1
Accommodation and Food Services	-0.1472	0.0587	-2.51
Female	-0.0752	0.0151	-4.99
Black	-0.1512	0.0282	-5.36
Asian	-0.0424	0.0230	-1.84
Native Hawaiian & Other Pacific Islander	-0.2013	0.1018	-1.98
American Indian & Alaskan Native	-0.0701	0.0539	-1.3
Other Race	-0.0129	0.0435	-0.3
Hispanic	0.0116	0.0250	0.46
Veteran	-0.2082	0.0256	-8.14
Not Classifiable	0.1012	0.0469	2.16
Franchise	-0.1145	0.0334	-3.42
Owner Age 25-34	0.1887	0.0285	6.61
Owner Age 35-44	0.2538	0.0285	8.92
Owner Age 45-54	0.2542	0.0286	8.89
Owner Age 55-64	0.1561	0.0303	5.16
Owner Age: 65 plus	0.0230	0.0370	0.62
Owner Education- Some College	0.0366	0.0193	1.89
Owner Education- College Degree	0.0999	0.0203	4.92
Owner Education- Graduate Degree	0.1568	0.0232	6.76
Previous Business Experience	0.0241	0.0157	1.54
Primary Source of Income	0.2016	0.0167	12.06
Hours Worked 20-39	0.1228	0.0204	6.02
Hours Worked 40	0.0569	0.0252	2.26
Hours Worked 41-59	0.2176	0.0232	9.38
Hours Worked 60 plus	0.2053	0.0245	8.37
Founded	-0.0155	0.0206	-0.75
Constant	-0.7217	0.0586	-12.31

N (unweighted) = 100,000
R2 = 0.0584

The second survival model includes only substantive firms: businesses with some financial capital investment, whose owners worked at least 20 hours a week in the business, and businesses founded by the current owner. This analysis, which can be found in Table 19, revealed some differences. The coefficients on all of the financial capital investment dummy variables were positive, but a few were no longer statistically significant. The coefficient on team ownership was now statistically significant and positively related to survival and the coefficient on home-based was no longer statistically significant. Two additional coefficients on industry dummies were now statistically significant, construction and other services. The coefficients on female, NHPI, and not classifiable were no longer statistically significant, however, the coefficient on Asian is now statistically significant and negative, indicating that for substantive businesses, Asian-owned businesses, like black-owned businesses, were less likely to survive than white-owned businesses.

The findings on owner age and education were similar to those for the full sample. However, the findings for hours worked was different and a bit unexpected. The coefficient on 40 hours a week was negative and statistically significant, indicating that businesses with owners working 40 hours a week were less likely to survive than businesses with owners working 20-39 hours a week, holding other factors constant. The other two coefficients on the greater hours worked variables were positive, but not statistically significant. While the coefficient wasn't large, this was still an unexpected result.

Table 19. Logistic Regression for Survival: Substantive Firms

	Coefficient	Std. err.	Z Statistic
Started as a Nonemployer	-0.5537	0.0280	-19.78
Corporation	0.4097	0.0301	13.6
Partnership	0.2048	0.0383	5.34
$5K-$10K FK investment	0.1372	0.0335	4.1

$10-$25K FK investment	0.0576	0.0331	1.74
$25-$50K FK investment	0.1015	0.0401	2.53
$50K-$100K FK investment	0.0711	0.0430	1.65
$100K-$250K FK investment	0.1008	0.0470	2.14
$250K-$1M FK investment	0.1415	0.0599	2.36
$1M+ FK investment	0.1842	0.0981	1.88
Team Ownership	-0.0742	0.0296	-2.5
Joint Ownership- Spouses	0.1824	0.0359	5.08
Intellectual Property	-0.0691	0.0375	-1.84
Has Business and/or Government Clients	0.0066	0.0230	0.28
Home based	-0.0332	0.0229	-1.45
Exporter	0.1796	0.0535	3.36
E-Commerce	0.0990	0.0335	2.96
Website	0.1704	0.0238	7.17
Construction	0.2052	0.0866	2.37
Manufacturing	0.1697	0.0904	1.88
Wholesale	0.1254	0.0962	1.3
Retail	0.0180	0.0847	0.21
Transportation & Warehousing	0.1361	0.0892	1.53
Information, Finance, & Insurance	0.1727	0.0881	1.96
Real Estate	0.4635	0.0936	4.95
Professional Services	0.0868	0.0831	1.05
Other Services	0.2567	0.0871	2.95
Administration & Support Services	0.0564	0.0895	0.63
Education Services	0.0924	0.1074	0.86
Health Care and Social Assistance	0.1670	0.0893	1.87
Arts, Entertainment, and Recreation	0.2378	0.0956	2.49
Accommodation and Food Services	-0.1469	0.0946	-1.55
Female	-0.0285	0.0234	-1.22
Black	-0.1498	0.0423	-3.54
Asian	-0.0962	0.0349	-2.76
Native Hawaiian and Other Pacific Islander	0.0821	0.1611	0.51
American Indian & Alaskan Native	-0.1585	0.0871	-1.82
Other Race	-0.0436	0.0654	-0.67
Hispanic	0.0276	0.0376	0.74
Veteran	-0.2135	0.0385	-5.54
Not Classifiable	-0.1861	0.3117	-0.6
Franchise	-0.0996	0.0533	-1.87
Owner Age 25-34	0.1807	0.0531	3.4
Owner Age 35-44	0.2258	0.0529	4.27
Owner Age 45-54	0.2686	0.0532	5.05

Owner Age 55-64	0.1484	0.0558	2.66
Owner Age: 65 plus	-0.0063	0.0681	-0.09
Owner Education- Some College	0.0380	0.0297	1.28
Owner Education- College Degree	0.1410	0.0315	4.47
Owner Education- Graduate Degree	0.1883	0.0362	5.2
Previous Business Experience	-0.0162	0.0224	-0.73
Primary Source of Income	0.2123	0.0240	8.85
Hours Worked 40	-0.1079	0.0322	-3.35
Hours Worked 41-59	0.0249	0.0290	0.86
Hours Worked 60 plus	0.0003	0.0304	0.01
Constant	-0.4333	0.1005	-4.31

N (unweighted) = 41,000
R2 = 0.0489

Positive Net Employment Growth

The final regression model explores the likelihood of businesses experiencing positive net employment growth over the observation period using logistic regression. The dependent variable is set to one for businesses with positive net employment growth and zero for those with zero or negative growth, or closure. The same set of control variables as in the previous models is used. The first set of regression results, which can be found in Table 20, makes use of the full CSP-SBO matched sample.

Starting as a nonemployer was negatively associated with experiencing positive net employment growth over the period. This was highly statistically significant. This finding aligns with broader literature that highlights the challenges faced by nonemployer firms in scaling up (Hipple & Hammond, 2016). Corporations and partnerships were much more likely to have positive net employment growth compared with sole proprietorships. Except for the less than $5,000 in financial capital investment, the rest of the coefficients on the financial capital

investment levels were all positive and statistically significant. This supports the notion that adequate capital is crucial for job creation and business expansion (Cassar, 2004).

Business characteristics that were positively associated with net positive employment growth were team ownership, joint ownership between spouses, and having a website. These findings highlight the importance of collaborative management structures (Colombo & Grilli, 2010) and digital engagement for growth (Bertschek, Cerquera, & Klein, 2013). On the other hand, intellectual property and home-based businesses are negatively associated with positive net employment growth, suggesting limitations in scaling operations from home settings. The only coefficient on an industry dummy that was statistically significant was a positive coefficient on health care and social assistance. These results were a bit surprising since previous research has highlighted industry-specific dynamics in job creation, with some sectors inherently better positioned to expand employment (Acs & Audretsch, 1990).

In terms of owner characteristics, businesses owned by women were less likely to have positive net employment growth than businesses owned by men. Black-owned businesses were less likely than white-owned businesses to have positive net employment growth. Veteran-owned businesses were less likely to have positive net employment growth than non-veteran-owned businesses. These findings underscore the need for targeted support to overcome systemic barriers faced by these groups (Fairlie & Robb, 2008). Companies not classified by race/ethnicity/sex were more likely to have positive net employment growth over the period than those that were private or classifiable. Owner age was important. For owners aged 25-64, their businesses were more likely to have positive net employment growth than businesses owned by owners that were younger than 25. Only the coefficient on the education dummy for some college was statistically

significant and it was negative, which indicates those businesses were less likely to have positive net employment growth than businesses with owners with a high school degree or less. Previous business experience and being a founder were not statistically significant. However, the primary source of income being from the business and hours worked are strongly positive and statistically significant factors, suggesting that practical experience and dedicated effort are critical for achieving positive employment outcomes (Lévesque & Minniti, 2006).

Table 20. Logistic Regression for Positive Net Employment Growth: All Firms

	Coefficient	Std. err.	Z Statistic
Started as a Nonemployer	-2.4760	0.0354	-69.92
Corporation	1.0420	0.0355	29.35
Partnership	0.8595	0.0421	20.41
Other Legal Form	1.7660	0.1167	15.14
< $5K FK investment	-0.0110	0.0356	-0.31
$5K-$10K FK investment	0.2521	0.0447	5.64
$10-$25K FK investment	0.1895	0.0409	4.63
$25-$50K FK investment	0.2227	0.0446	4.99
$50K-$100K FK investment	0.2877	0.0441	6.53
$100K-$250K FK investment	0.3323	0.0440	7.56
$250K-$1M FK investment	0.3768	0.0504	7.48
$1M+ FK investment	0.2825	0.0693	4.08
Team Ownership	0.1094	0.0289	3.79
Joint Ownership- Spouses	0.0685	0.0360	1.9
Intellectual Property	-0.1093	0.0372	-2.94
Has Business and/or Government Clients	0.0275	0.0248	1.11
Home based	-0.2933	0.0275	-10.66
Exporter	0.0100	0.0539	0.18
E-Commerce	0.0123	0.0346	0.36
Website	0.3999	0.0245	16.34
Construction	0.3609	0.0996	3.62
Manufacturing	0.1739	0.1011	1.72
Wholesale	0.1727	0.1058	1.63
Retail	0.0189	0.0962	0.2
Transportation & Warehousing	-0.0057	0.1026	-0.06
Information, Finance, & Insurance	-0.0100	0.0998	-0.1
Real Estate	-0.0632	0.1047	-0.6
Professional Services	-0.0204	0.0950	-0.21

Other Services	0.0314	0.1011	0.31
Administration & Support Services	0.0662	0.1029	0.64
Education Services	0.1770	0.1160	1.53
Health Care and Social Assistance	0.4019	0.1001	4.01
Arts, Entertainment, and Recreation	-0.1824	0.1104	-1.65
Accommodation and Food Services	0.0098	0.1009	0.1
Female	-0.0887	0.0257	-3.45
Black	-0.2909	0.0585	-4.98
Asian	-0.0491	0.0360	-1.36
Native Hawaiian & Other Pacific Islander	-0.1444	0.2800	-0.52
American Indian & Alaskan Native	-0.2160	0.1194	-1.81
Other Race	-0.0142	0.0805	-0.18
Hispanic	0.0763	0.0443	1.72
Veteran	-0.1691	0.0435	-3.89
Not Classifiable	0.2930	0.0816	3.59
Franchise	0.0168	0.0462	0.36
Owner Age 25-34	0.4163	0.0701	5.94
Owner Age 35-44	0.4284	0.0685	6.26
Owner Age 45-54	0.3263	0.0685	4.76
Owner Age 55-64	0.1576	0.0707	2.23
Owner Age: 65 plus	0.0474	0.0805	0.59
Owner Education- Some College	-0.0930	0.0337	-2.76
Owner Education- College Degree	0.0204	0.0337	0.61
Owner Education- Graduate Degree	0.0565	0.0382	1.48
Previous Business Experience	0.0369	0.0239	1.55
Primary Source of Income	0.1606	0.0282	5.69
Hours Worked 20-39	0.3476	0.0395	8.79
Hours Worked 40	0.4524	0.0426	10.62
Hours Worked 41-59	0.5840	0.0385	15.16
Hours Worked 60 plus	0.6614	0.0395	16.76
Founded	-0.0197	0.0311	-0.64
Constant	-2.4310	0.1159	-20.97

N (unweighted) = 100,000
R2 = 0.3556

The final logistic model is for the subsample of substantive firms, which yielded similar results with a few important exceptions. The results are found in Table 21. The coefficient on team ownership is no longer statistically significant and the coefficient on having business and/or

government clients is now statistically significant (and positively associated with positive net employment growth). Two additional industry dummies are now statistically significant (and positive): manufacturing and education services. The coefficients on Asian and American Indian & Alaskan Native are now statistically significant and negative, indicating that they are less likely than white-owned businesses to have positive net employment growth. Finally, the coefficient on the 55-64 age category is no longer statistically significant.

Table 21. Logistic Regression for Positive Net Employment Growth: Substantive Firms

	Coefficient	Std. err.	Z Statistic
Started as a Nonemployer	-2.2140	0.0431	-51.33
Corporation	0.9925	0.0436	22.77
Partnership	0.8379	0.0537	15.6
$5K-$10K FK investment	0.1863	0.0472	3.95
$10-$25K FK investment	0.1354	0.0445	3.04
$25-$50K FK investment	0.2232	0.0503	4.44
$50K-$100K FK investment	0.2289	0.0518	4.42
$100K-$250K FK investment	0.2642	0.0548	4.82
$250K-$1M FK investment	0.3017	0.0675	4.47
$1M+ FK investment	0.1196	0.1061	1.13
Team Ownership	0.0204	0.0372	0.55
Joint Ownership- Spouses	0.1024	0.0460	2.22
Intellectual Property	-0.1436	0.0477	-3.01
Has Business and/or Government Clients	0.0624	0.0319	1.96
Home based	-0.3235	0.0331	-9.76
Exporter	0.0787	0.0696	1.13
E-Commerce	-0.0487	0.0450	-1.08
Website	0.3730	0.0317	11.78
Construction	0.4981	0.1338	3.72
Manufacturing	0.3057	0.1383	2.21
Wholesale	0.2655	0.1439	1.85
Retail	0.0852	0.1317	0.65
Transportation & Warehousing	0.1546	0.1376	1.12
Information, Finance, & Insurance	0.0687	0.1357	0.51
Real Estate	-0.0172	0.1450	-0.12
Professional Services	0.1070	0.1295	0.83
Other Services	0.1262	0.1382	0.91
Administration & Support Services	0.2372	0.1391	1.71

Education Services	0.4218	0.1582	2.67
Health Care and Social Assistance	0.5731	0.1370	4.18
Arts, Entertainment, and Recreation	-0.1814	0.1533	-1.18
Accommodation and Food Services	0.0546	0.1397	0.39
Female	-0.0710	0.0326	-2.18
Black	-0.2468	0.0682	-3.62
Asian	-0.1102	0.0466	-2.37
Native Hawaiian & Other Pacific Islander	0.0743	0.3160	0.24
American Indian & Alaskan Native	-0.2919	0.1480	-1.97
Other Race	0.0407	0.0972	0.42
Hispanic	0.0638	0.0540	1.18
Veteran	-0.1963	0.0549	-3.58
Not Classifiable	0.5155	0.5526	0.93
Franchise	0.0093	0.0684	0.14
Owner Age 25-34	0.4098	0.1100	3.73
Owner Age 35-44	0.3883	0.1088	3.57
Owner Age 45-54	0.2902	0.1092	2.66
Owner Age 55-64	0.0985	0.1121	0.88
Owner Age: 65 plus	-0.0893	0.1264	-0.71
Owner Education- Some College	-0.0925	0.0430	-2.15
Owner Education- College Degree	0.0752	0.0438	1.72
Owner Education- Graduate Degree	0.0791	0.0496	1.6
Previous Business Experience	-0.0222	0.0295	-0.75
Primary Source of Income	0.1639	0.0344	4.76
Hours Worked 40	0.1077	0.0481	2.24
Hours Worked 41-59	0.2464	0.0426	5.78
Hours Worked 60 plus	0.3349	0.0436	7.69
Constant	-2.0740	0.1734	-11.96

N (unweighted) = 41,000
R2 = 0.28

The logistic regression results provide several key insights into factors that influence positive net employment growth, which have policy implications. Our results suggest that programs focusing on providing additional support to nonemployer businesses could effectively enhance their capacity for employment growth. We show evidence that enhancing access to financial capital can be crucial. Our results suggest that programs aimed at reducing capital constraints can support job creation and business expansion and the support of digital adoption and website

presence can help businesses engage with customers and scale operations more effectively. Tailored support for different industries can address unique challenges and opportunities, fostering employment growth where it is most viable. Targeted support for female, minority, and veteran entrepreneurs can help overcome systemic barriers to growth. Promoting practical experience and dedicated work effort can significantly impact employment outcomes, suggesting a balanced approach to education and hands-on business management.

Conclusions

Understanding the dynamics of startup business survival and growth is crucial for fostering a more entrepreneurial economy. Job creation remains a critical issue for policymakers, researchers, and advocacy groups. This research sought to shed light on the transition from non-employer businesses to employer businesses and job creation by new startups. Despite the significance of this transition, researchers are only beginning to have a better understanding of which nonemployer companies become employer businesses and the process they undergo. Using the newly available Comprehensive Startup Panel (CSP) (2011-2019) and the 2012 Survey of Business Owners (SBO), we track startups that began operations in 2011 or 2012—both with and without employees—over the 2012-2019 period. This enables us to understand better the transition from non-employer to employer status, firm survival, and job creation by startups. The matched CSP-SBO sample offers a unique opportunity to examine a large set of more than 100,000 startup businesses tracked over their first seven years, providing detailed information on the firms, owners, and strategy.

We find strong evidence that the legal form is related to the likelihood of transitioning to an employer firm with corporations and partnerships more likely to transition than their sole proprietor counterparts. We also find financial capital investment is another critical factor influencing the likelihood of transition with higher levels of investment being correlated with the likelihood to transition. Other firm characteristics that were positively associated with transitioning from nonemployer to employer were joint ownership between spouses, having a website, and having customers that were other business and/or government. Owner demographics, such as sex, race, and ethnicity, generally do not significantly influence the transition. However, being black is negatively associated with the likelihood of transitioning, while companies not classifiable by race/gender/ethnicity are more likely to transition. Education levels surprisingly did not appear to be correlated with the likelihood of transitioning, however previous business experience had a positive and statistically significant effect, as did having a business that provides the primary source of income. Business owners ages 25 to 44 are more likely to transition their businesses compared with businesses with the youngest owners (under 25), however transitions for business with owners aged 45 or more were not statistically different from businesses with the youngest owners. Hours worked per week is a significant positive factor; more hours worked correlates with a higher likelihood of transitioning, emphasizing the importance of effort and dedication in moving from nonemployer to employer status.

Regarding business survival, businesses starting as a nonemployers are less likely than employer startups to survive. Legal form strongly correlates with survival: corporations, partnerships, and other legal forms are more likely to survive than sole proprietorships. Higher levels of

financial capital investment are significantly associated with higher survival rates. Firm characteristics positively influencing survival include joint ownership between spouses, exporting goods or services, having e-commerce capabilities, and maintaining a website. The varied and significant coefficients across industry dummy variables underscore the importance of industry in determining survival likelihood.

In terms of owner characteristics, businesses owned by women were less likely to survive than businesses owned by men. Businesses owned by blacks and Native Hawaiian and Other Pacific Islanders (NHOPI) were less likely to survive than businesses owned by whites. Veteran-owned businesses were less likely to survive than businesses owned by non-veterans. Businesses that were not classifiable by race/ethnicity/sex were more likely to survive than private businesses and those that were classifiable. Owner age, education, and hours worked were all important predictors in business survival. The coefficients on all owner age categories were positive and statistically significant except the one for 65 years and older, which indicates that the youngest owners (under 25 years old) were the least likely to survive. In terms of education, owners with a college degree or graduate degree were more likely to survive than businesses with owners with only a high school education or less. Businesses with owners working more hours than 20 hours a week were more likely to survive than businesses with owners working less than 20 hours a week. Finally, businesses that were the primary source of income for the owner were more likely to survive than businesses that were not the primary source of income.

If the Small Business Administration (SBA) wanted to expand programs targeting firms at high risk of closure to improve their odds of survival, our results suggest these include women-led companies, businesses owned by blacks and Native Hawaiian and Other Pacific

Islanders (NHOPI), veterans, and young entrepreneurs (under 25 years old). SBA currently focuses on capital access as a primary policy concern. Our results support this approach, as we find that adequate financial capital is a crucial factor in increasing the survival prospects of young firms. Our results suggest that educating entrepreneurs on the importance of focusing their time and energy on the business and ensuring the business has the opportunity for a digital presence can also support the successful paths of entrepreneurs.

Our results also provide several key insights into factors that influence positive net employment growth thereby informing potential policy approaches. We find that access to financial capital is crucial, so policies aimed at reducing capital constraints can support job creation and business expansion. Supporting a website presence can help businesses engage with customers. Targeted support for female, Black, and veteran entrepreneurs can help overcome systemic barriers to growth. Promoting dedicated work effort can significantly impact employment outcomes.

Our analysis of a large sample of startup businesses tracked over seven years post-startup provides key insights into factors influencing both survival and growth. Surprisingly, education was not strongly correlated with the dynamics as hypothesized although having a college degree or graduate degree was important in our survival analysis. Business strategies--- dedicating more time and effort into the business in terms of hours worked, creating a business that is the primary source of income, establishing a digital presence, building a business outside of the home, and investing sufficient financial capital in the business to support the operations and growth appear to be the strongest factors in positive outcomes for business startups. As the SBA aims to be more impactful in its support of early-stage companies, we hope our findings offer some guidance on focusing efforts to be most effective.

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